

Introduction to logarithms:

Given the equation $2^x = 32$, how would you find the value of x ?

$$x = \log_2 32 = 5$$

$$a^x = b$$

$$x = \log_a b$$

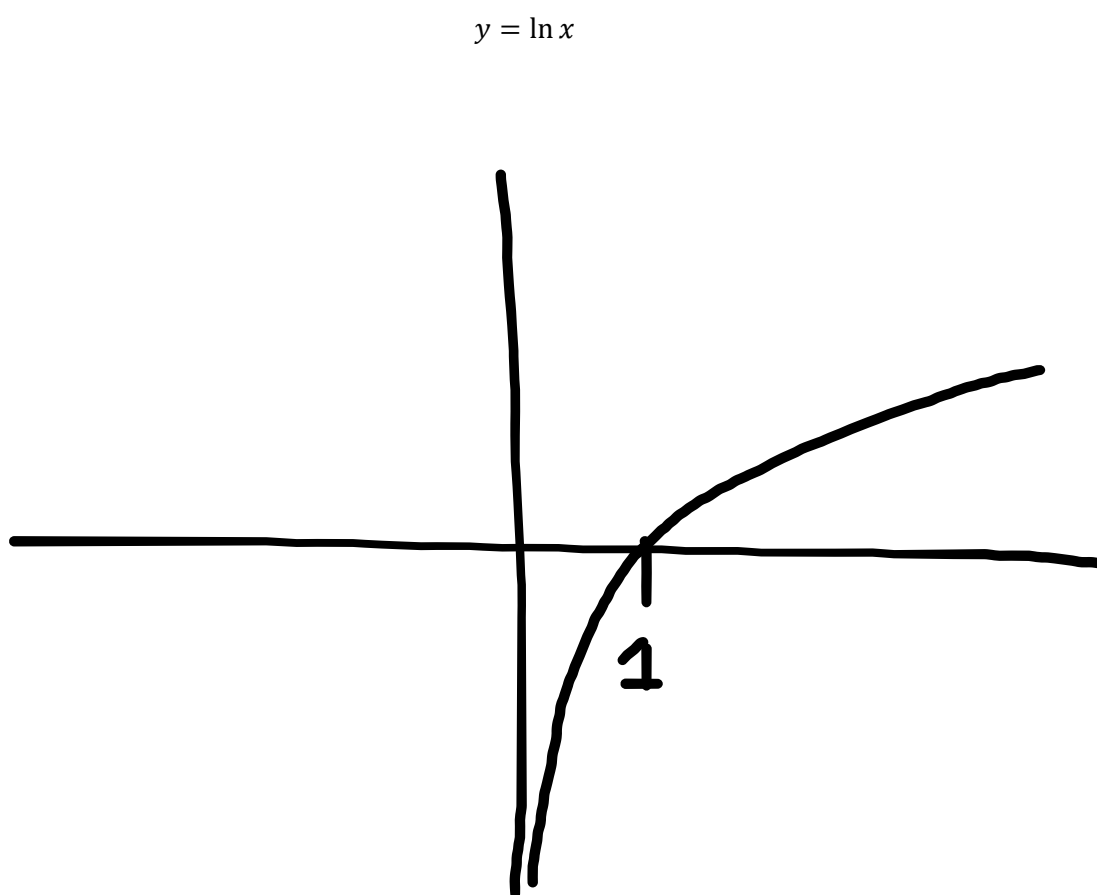
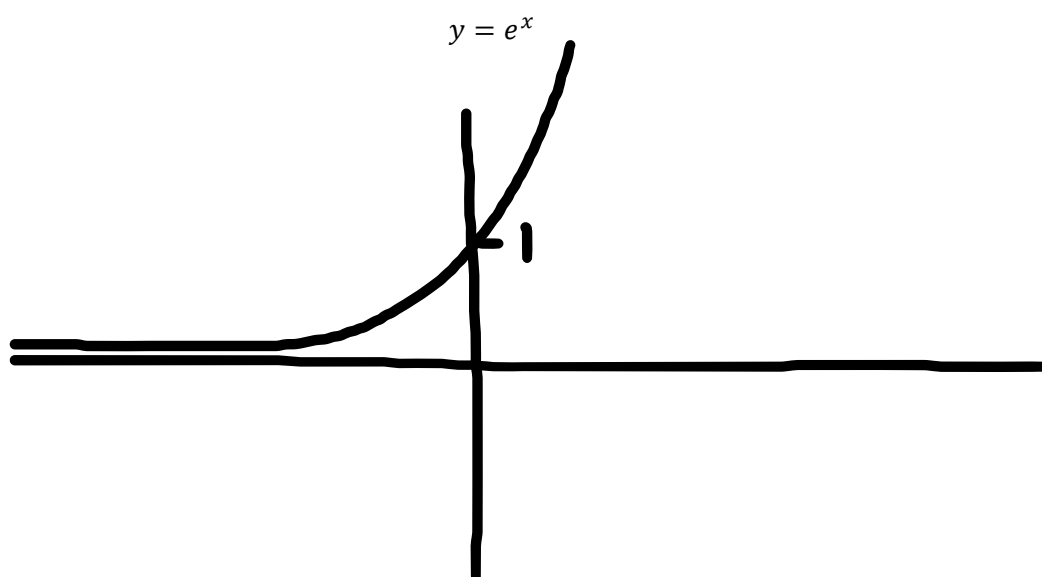
Natural logarithms:

e is a special irrational number similar to π .

$$y = e^x$$

$$x = \log_e y = \ln y$$

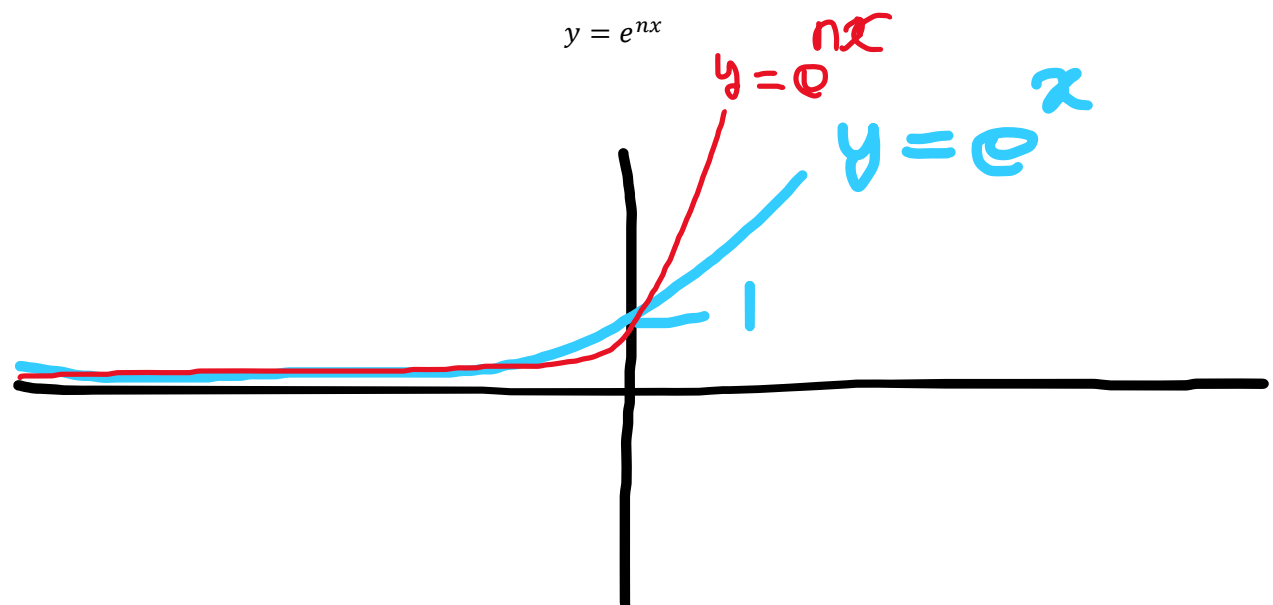
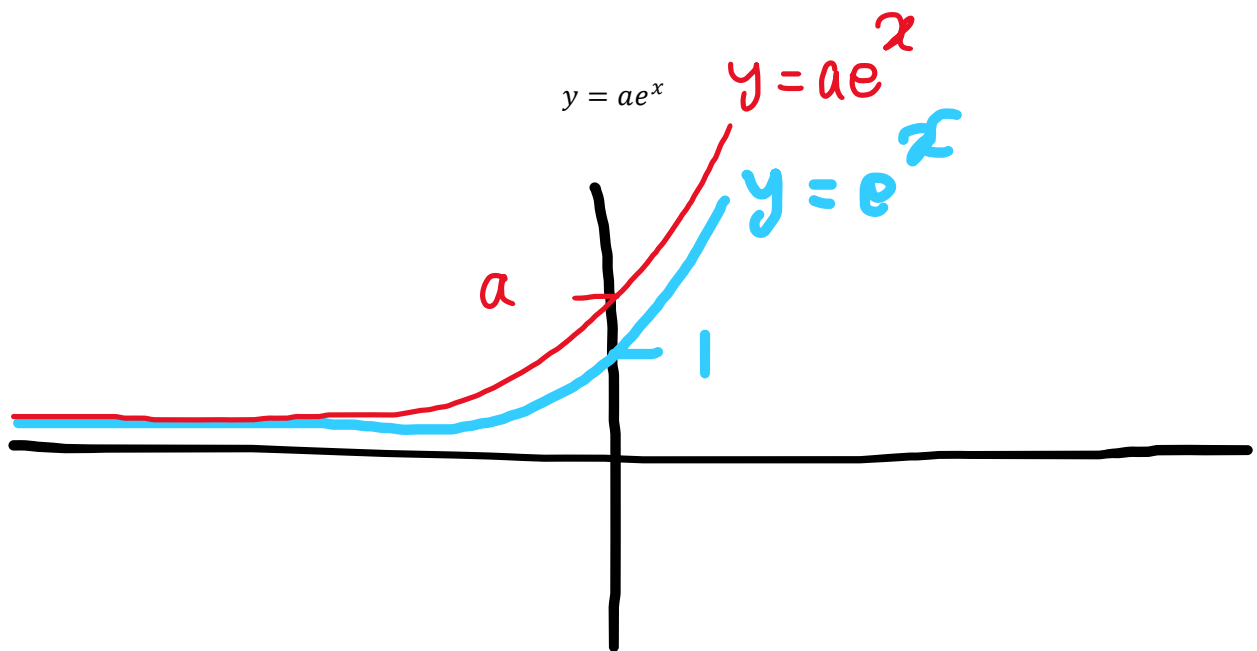
$f(x) = e^x$ and $g(x) = \ln x$ are inverse functions of each other

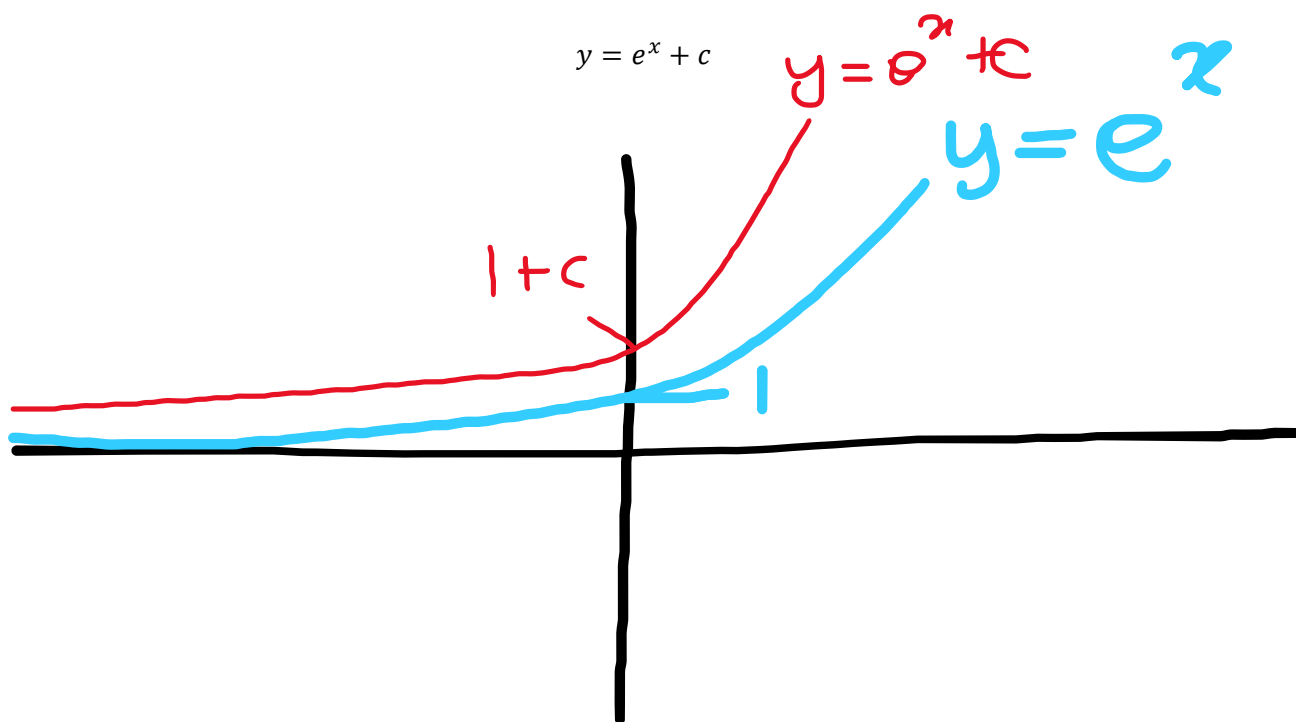


The graph of $y = e^x$ has a horizontal asymptote, the line $y = 0$ (or the x-axis).

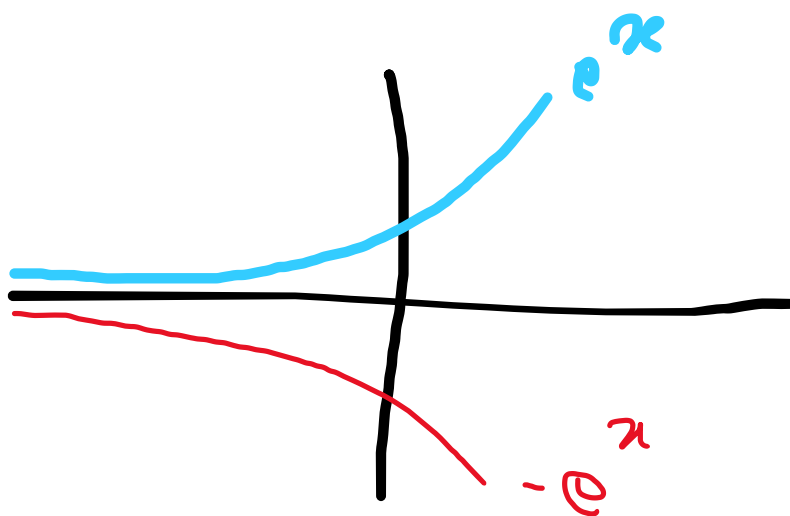
The graph of $y = \ln x$ has a vertical asymptote, the line $x = 0$ (or the y-axis).

Asymptotes are lines that graphs tend towards but never intersect.

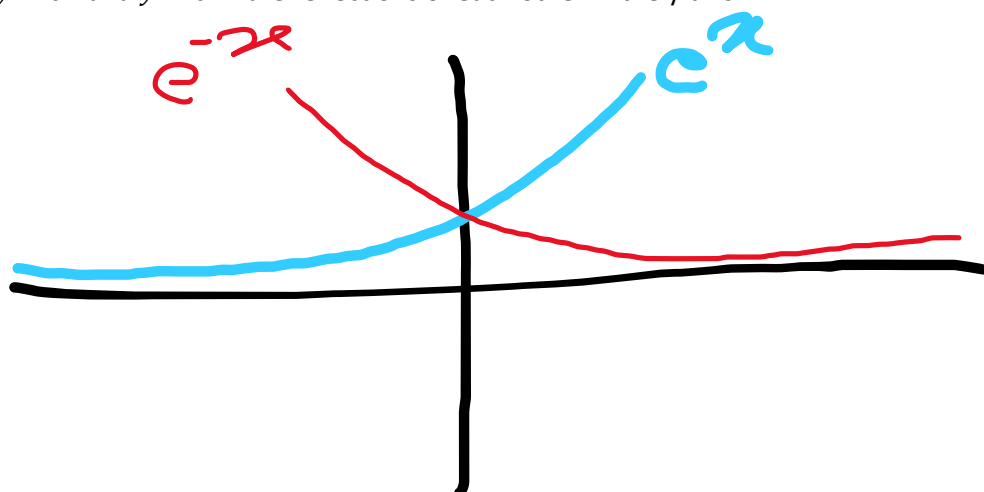




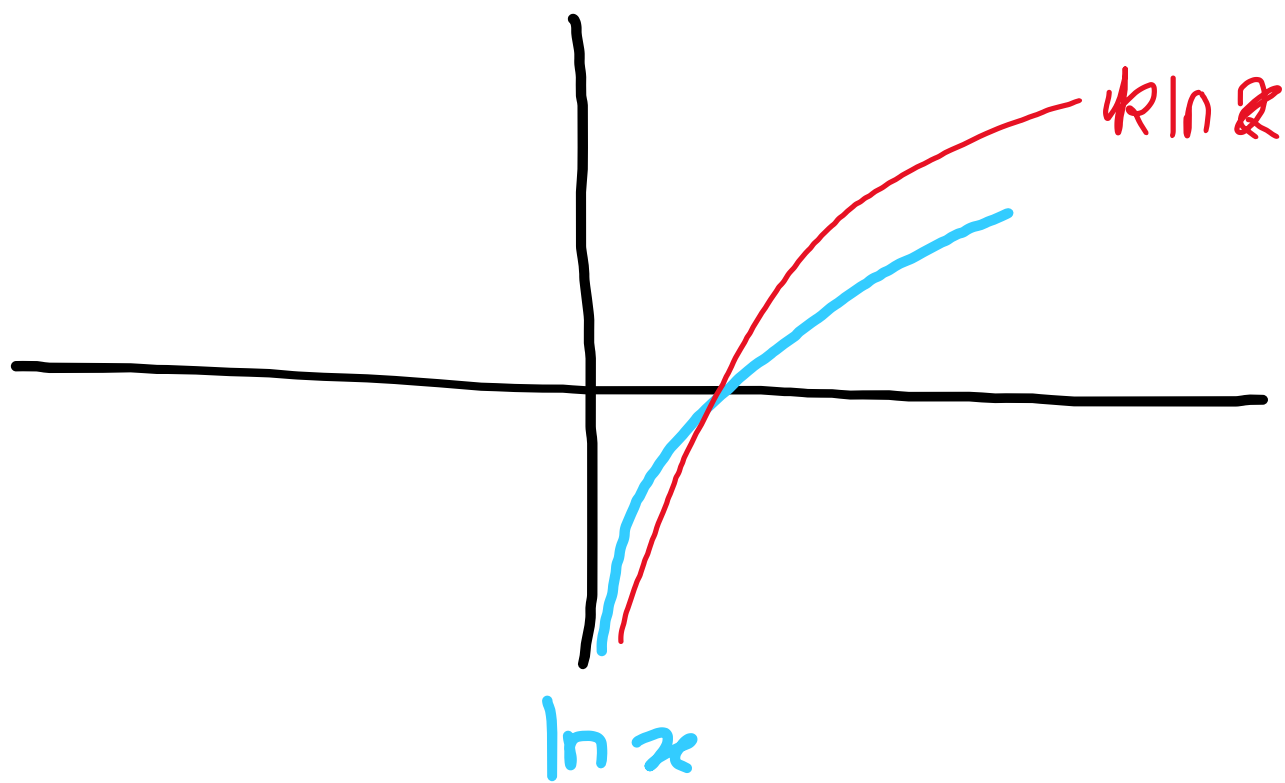
The graphs of $y = e^x$ and $y = -e^x$ are reflections of each other in the x-axis



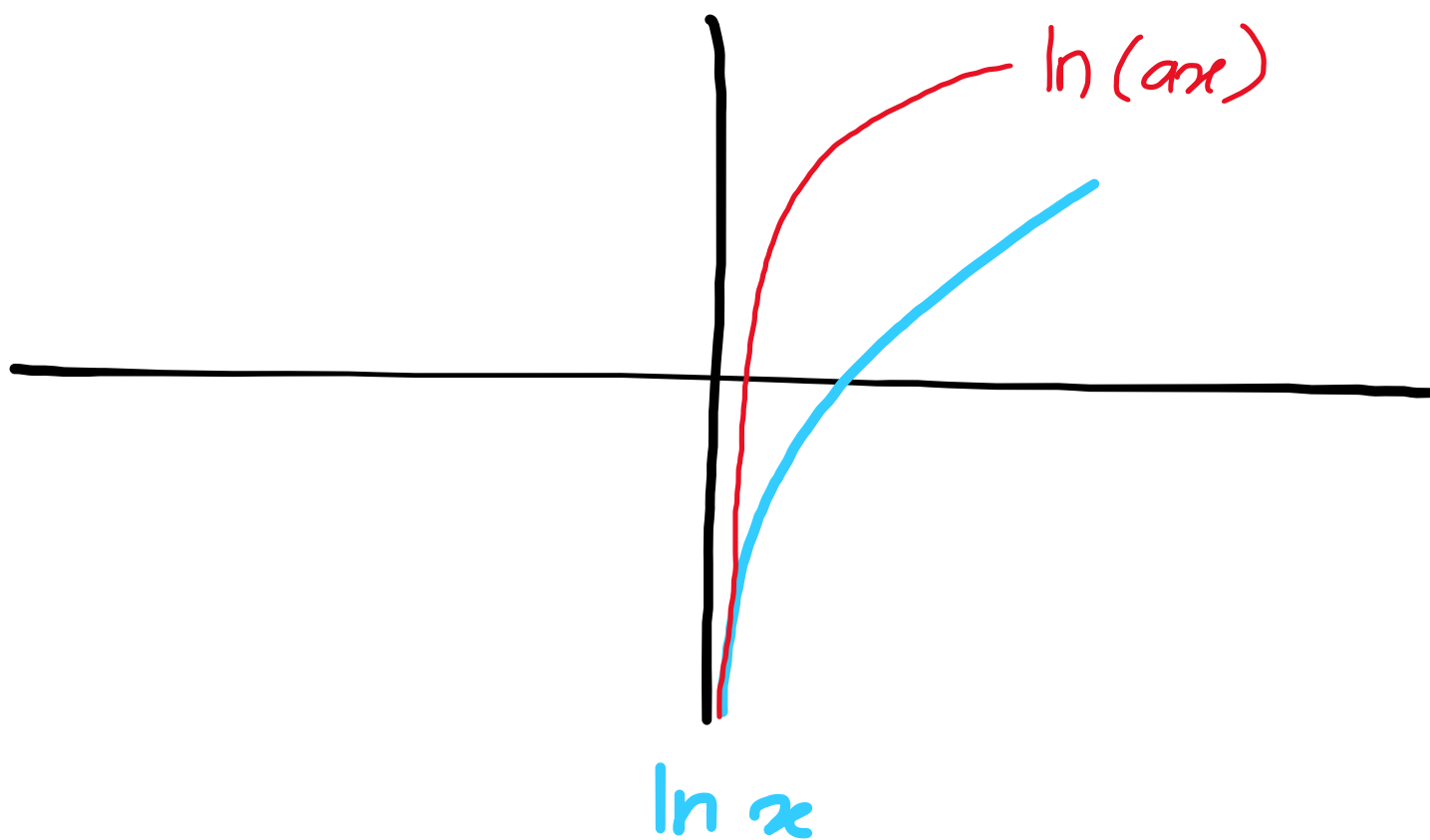
The graphs of $y = e^x$ and $y = e^{-x}$ are reflections of each other in the y-axis

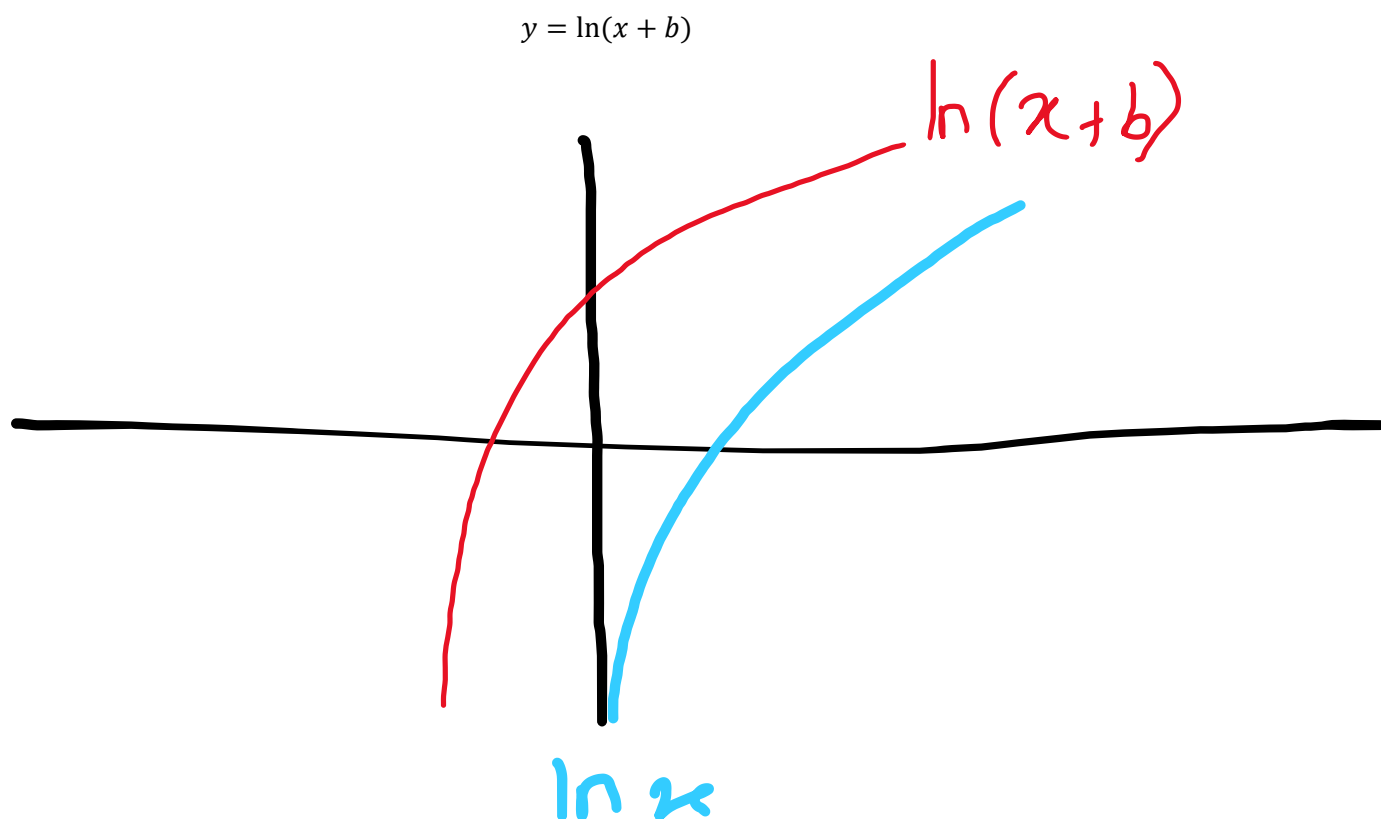


$$y = k \ln x$$



$$y = \ln(ax)$$





Reflections of the graphs of $y = \ln x$ are similar to those of $y = e^x$

Laws of logarithms:

$$a^x = b$$
$$x = \log_a b$$

$$10^x = a$$
$$x = \log_{10} a = \lg a$$

$$\log_a a = 1$$

$$\log_a b^c = c \log_a b$$

$$\log_a 1 = 0$$

$$\log_a(bc) = \log_a b + \log_a c$$

$$\log_a \left(\frac{b}{c} \right) = \log_a b - \log_a c$$

$$a^{\log_a b} = b$$

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Example:

$$3 + 2 \lg p - \lg q = 3 \lg 10 + \lg p^2 - \lg q = \lg 1000 + \lg \frac{p^2}{q} = \lg \frac{1000p^2}{q}$$

$$\frac{1}{\log_5 e} = \frac{1}{\frac{\log_e e}{\log_e 5}} = \frac{\ln 5}{1} = \ln 5$$